

ASUNCION, Paraguay

WMO: 862180

Lat: 25.240S

Lon: 57.519W

Elev: 292

StdP: 14.54

Time Zone: -4.00 (PRY)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.4	44.9	31.7	26.4	54.6	35.4	30.6	53.4	26.1	66.8	23.4	67.8	2.5	180	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.2	99.0	75.4	97.1	75.6	95.3	75.6	80.5	91.2	79.7	90.2	79.0	89.2	13.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.6	145.1	85.3	77.0	142.1	85.0	75.7	136.1	83.9	44.5	90.4	43.6	91.8	42.6	88.5	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.9	18.8	DB	35.5	102.9	3.0	1.9	33.3	104.2	31.6	105.4	29.9	106.4	27.8	107.8	(4)
(5)			WB	34.5	82.9	2.9	1.3	32.4	83.8	30.7	84.6	29.0	85.3	26.9	86.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.4	83.1	81.8	79.5	74.9	67.8	65.5	65.0	68.5	71.6	76.3	78.0	81.2	(6)	
	DBStd	9.58	3.84	4.56	4.99	6.81	7.39	8.86	9.80	10.17	9.24	6.88	5.51	4.65	(7)	
	HDD50	8	0	0	0	0	0	2	5	1	0	0	0	0	(8)	
	HDD65	432	0	0	0	10	57	106	131	85	40	3	0	0	(9)	
	CDD50	8919	1028	891	916	747	552	468	471	576	647	815	840	968	(10)	
	CDD65	3870	563	471	451	308	144	122	131	195	237	354	391	503	(11)	
	CDH74	39566	6609	5174	4564	2675	956	724	935	1910	2482	3641	4130	5766	(12)	
	CDH80	18106	3347	2506	2088	1078	268	170	280	794	1126	1681	1911	2857	(13)	
(14) Wind	WSAvg	8.0	7.3	7.1	6.6	6.9	7.0	7.8	8.6	9.4	9.6	9.4	8.4	7.8	(14)	
(15) Precipitation	PrecAvg	56.8	4.8	6.3	5.2	6.8	4.6	2.6	2.1	1.6	3.0	5.6	7.0	7.2	(15)	
	PrecMax	89.1	10.3	17.3	10.8	12.9	14.6	6.5	6.6	9.1	7.9	12.5	19.3	18.5	(16)	
	PrecMin	41.4	0.6	1.2	0.9	1.3	0.1	0.0	0.0	0.0	0.2	0.8	0.7	1.7	(17)	
	PrecStd	12.2	2.8	3.8	2.6	3.0	3.5	1.8	1.8	2.3	2.2	3.2	4.2	3.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.5	102.1	98.7	95.7	89.8	87.5	89.6	95.5	99.2	102.0	100.2	100.0	(19)	
		MCWB	76.3	75.4	75.4	76.3	73.1	70.7	69.4	69.6	71.9	75.7	76.3	75.5	(20)	
	2%	DB	98.3	97.3	95.4	92.9	86.3	84.4	86.2	91.8	95.3	96.7	96.4	96.9	(21)	
		MCWB	76.0	76.2	76.1	76.0	72.9	70.5	69.1	69.5	71.6	75.4	75.9	76.5	(22)	
	5%	DB	96.0	95.0	93.1	89.8	82.8	82.0	83.4	88.2	91.2	92.9	93.1	94.9	(23)	
		MCWB	76.0	76.5	75.7	75.0	71.6	70.3	68.4	68.4	71.0	74.8	74.7	76.6	(24)	
	10%	DB	93.5	91.8	90.0	86.3	79.1	78.5	79.2	84.3	86.4	89.2	89.7	92.5	(25)	
		MCWB	75.9	76.3	75.0	74.0	70.0	69.3	67.0	67.3	69.2	73.4	73.4	76.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.5	81.6	80.2	80.1	77.3	74.3	74.0	74.1	77.2	79.9	80.3	81.8	(27)	
		MCDB	92.1	92.0	90.1	90.4	84.2	81.9	83.2	89.1	90.7	93.1	92.0	92.3	(28)	
	2%	WB	80.1	80.0	79.2	78.3	75.1	72.6	72.0	72.0	74.9	78.0	78.9	80.0	(29)	
		MCDB	90.6	90.0	89.0	87.4	82.2	80.1	80.9	86.1	88.0	90.9	90.8	90.5	(30)	
	5%	WB	79.3	79.0	78.0	76.9	73.1	71.2	70.0	70.1	72.7	76.3	77.4	79.1	(31)	
		MCDB	89.5	89.0	87.6	85.5	80.2	78.5	79.0	83.3	86.2	88.7	88.3	89.2	(32)	
	10%	WB	78.3	78.0	77.0	75.6	71.3	69.8	68.1	68.4	71.0	74.7	75.9	78.0	(33)	
		MCDB	88.0	87.4	86.2	83.5	77.7	77.1	77.3	80.8	83.2	86.2	85.7	87.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	18.3	18.6	17.8	16.4	16.2	18.3	20.5	20.3	18.8	19.4	19.1	(35)	
		MCDBR	22.5	22.2	22.1	20.9	18.8	17.8	20.0	22.7	24.1	23.0	23.2	22.2	(36)	
	5% WB	MCWBR	7.0	6.9	7.7	8.3	9.0	8.6	9.4	9.6	10.1	9.4	9.1	8.0	(37)	
		MCWBR	18.5	18.2	18.5	18.1	16.4	16.0	18.3	21.4	21.2	20.4	19.3	18.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.411	0.413	0.401	0.400	0.380	0.403	0.397	0.505	0.586	0.504	0.414	0.417	(40)	
		taud	2.403	2.405	2.419	2.400	2.418	2.346	2.348	1.995	1.802	2.087	2.352	2.373	(41)	
	Ebn at Noon		296	291	285	270	261	246	253	234	229	262	293	295	(42)	
		Edh at Noon	40	39	37	35	31	32	33	51	68	54	42	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		2130	1939	1715	1399	1045	878	1020	1222	1454	1714	2009	2133	(44)	
	RadStd		141	143	122	143	142	90	102	87	157	182	120	131	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.21	+0.56	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	+0.92	+0.36	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page